

Universality and the right-angled Coxeter envelope of orthoscheme reflection groups

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Abstract

The right-triangle orbit-growth universality of the companion papers is the planar case of a general phenomenon. We prove an abstract dichotomy: for any family of quotients of a common group, spherical growth is shape-independent through a horizon equal to half the shortest added relator and first deviates exactly there. For orthoscheme (path-simplex) reflection groups the common group is the *right-angled Coxeter group* (RACG) on the orthoscheme's orthogonality graph, whose growth we compute in closed form: it is rational, with numerator $(1+t)^{\lfloor n/2 \rfloor + 1}$ and growth rate the algebraic number $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$. The geometric group $W_T \leq \text{Isom}(\mathbb{R}^n)$ is a quotient of this RACG, and the dichotomy gives an exact alternative: either the surjection $\rho: W_n \twoheadrightarrow W_T$ is injective, in which case the geometric growth is the rational envelope in every degree, or it is not, in which case the growth agrees with the envelope up to a finite horizon and drops strictly below it there. In the plane the second alternative holds, with horizon 10, where the growth becomes the sequence A396406. For $n \geq 3$ we show that the amenability argument that settles the plane does *not* extend, because $O(n)$ contains a free subgroup of rank two as an abstract group, and we leave the alternative open in both directions.

1 The abstract universality–deviation dichotomy

Let $G = \langle S \rangle$ be a group with finite symmetric generating set S and word length ℓ_G . For $N \trianglelefteq G$ write $Q = G/N$, $s_d(H) = \#\{h : \ell_H(h) = d\}$, and $K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$, with $K(\{1\}) = \infty$.

Theorem 1 (Universality–deviation dichotomy). *Let $N \neq \{1\}$ and put $n := \lceil K(N)/2 \rceil$. Then $s_d(Q) = s_d(G)$ for $d < n$, and $s_n(Q) < s_n(G)$.*

Proof. Write $\pi : G \rightarrow Q$ and $K = K(N)$.

Agreement below n . Let $d < n$. If $d \geq 1$ then $2d \leq 2n - 2 \leq K - 1 < K$, since $2n \leq K + 1$. Suppose $g, h \in B_d(G)$ satisfy $\pi g = \pi h$. Then $gh^{-1} \in N$ and $\ell_G(gh^{-1}) \leq 2d < K$, so $gh^{-1} = 1$ and $g = h$. Thus π is injective on $B_d(G)$. It is also length-preserving there: if $\ell_G(g) = d$ and $\ell_Q(\pi g) = e < d$, pick h with $\ell_G(h) = e$ and $\pi h = \pi g$; then $\ell_G(gh^{-1}) \leq d + e < 2d < K$ forces $g = h$, contradicting $e < d$. Finally π maps the sphere of radius d in G onto the sphere of radius d in Q , because $\ell_Q(q) = \min\{\ell_G(g) : \pi g = q\}$ realises every q with $\ell_Q(q) = d$ by some g of length exactly d . Hence $s_d(Q) = s_d(G)$.

Strict drop at n . Choose $g_0 \in N \setminus \{1\}$ with $\ell_G(g_0) = K$ and write a geodesic word for it as $g_0 = w_1 w_2$ with $\ell_G(w_1) = n$ and $\ell_G(w_2) = K - n = \lfloor K/2 \rfloor$; both factors are geodesic as written. From $w_1 w_2 \in N$ we get $\pi(w_1) = \pi(w_2^{-1})$, and $w_1 \neq w_2^{-1}$ since otherwise $g_0 = 1$. Surjectivity of π from the radius- n sphere of G onto that of Q , established in the previous paragraph for $d < n$, holds verbatim for $d = n$.

If K is even then $n = K/2 = \lfloor K/2 \rfloor$, so w_1 and w_2^{-1} are two distinct elements of the sphere of radius n in G with the same image, and $s_n(Q) \leq s_n(G) - 1$.

If K is odd then $n = (K+1)/2$ and $\ell_G(w_2^{-1}) = n-1$, so $\ell_Q(\pi w_1) \leq n-1 < n$ and the image of w_1 does not lie on the sphere of radius n in Q . That sphere is therefore covered by π applied to the radius- n sphere of G with w_1 removed, and again $s_n(Q) \leq s_n(G) - 1$. $\square$

Applied to a family $\{Q_T = G/N_T\}$, this is universality: all members share G 's growth through their horizon $n_T = \lceil K(N_T)/2 \rceil$, then deviate. For orthoscheme reflection groups the ambient G is a rational-growth RACG, so the shared rational growth is an envelope whose validity is controlled entirely by whether N_T is trivial.

2 The orthoscheme family and its RACG envelope

An n -orthoscheme, or path simplex, has vertices $P_0, \dots, P_n$ with mutually orthogonal consecutive edges $P_{k-1}P_k = \ell_k e_k$; its $n+1$ facet reflections $R_0, \dots, R_n$ generate $W_T \leq \text{Isom}(\mathbb{R}^n)$ (the right-triangle group when $n=2$). Orthoschemes are the simplices for which the reflection group generated by the facets was first analysed systematically [2]. We normalise $\ell_1 = 1$ and write $U = \{\ell = (\ell_1, \dots, \ell_n) : \ell_i > 0\}$ for the leg space.

Lemma 2 (Facet normals). *Take $P_0 = 0$, so $P_k = \sum_{i \leq k} \ell_i e_i$. For every $\ell \in U$ the facet of the orthoscheme opposite P_j has normal*

$$m_0 = e_1, \quad m_j = \ell_{j+1} e_j - \ell_j e_{j+1} \quad (1 \leq j \leq n-1), \quad m_n = e_n.$$

Consequently $\text{supp}(m_0) = \{1\}$, $\text{supp}(m_j) = \{j, j+1\}$ and $\text{supp}(m_n) = \{n\}$, so that $m_i \cdot m_j = 0$ whenever $|i-j| \geq 2$, while $m_i \cdot m_{i+1} \neq 0$ for every i .

Proof. The facet opposite P_j is the affine hull of $\{P_i : i \neq j\}$. For $j=0$ the difference vectors $P_{k+1} - P_k = \ell_{k+1} e_{k+1}$, $1 \leq k \leq n-1$, span $\langle e_2, \dots, e_n \rangle$, whence $m_0 = e_1$. For $j=n$ the vertices $P_0, \dots, P_{n-1}$ span $\langle e_1, \dots, e_{n-1} \rangle$, whence $m_n = e_n$. For $1 \leq j \leq n-1$ the difference vectors are $\ell_k e_k$ for $k \leq j-1$, then $P_{j+1} - P_{j-1} = \ell_j e_j + \ell_{j+1} e_{j+1}$, then $\ell_k e_k$ for $k \geq j+2$. Orthogonality to the first and third families confines m_j to coordinates $\{j, j+1\}$, and orthogonality to the middle vector then forces $m_j \propto \ell_{j+1} e_j - \ell_j e_{j+1}$. The support statement is immediate, and two supports of the listed form are disjoint exactly when $|i-j| \geq 2$. For adjacent indices, $m_0 \cdot m_1 = \ell_2$, $m_j \cdot m_{j+1} = -\ell_j \ell_{j+2}$ for $1 \leq j \leq n-2$, and $m_{n-1} \cdot m_n = -\ell_{n-1}$, all nonzero because every $\ell_i > 0$. $\square$

Lemma 2 holds for *all* legs, with no genericity hypothesis and no case check: $R_i^2 = 1$ always, and $R_i R_j = R_j R_i$ whenever $|i-j| \geq 2$. For generic legs consecutive facets meet at an irrational multiple of π , imposing no finite bond. The largest group compatible with these relations is the right-angled Coxeter group [3]

$$W_n = \langle R_0, \dots, R_n \mid R_i^2 = 1, (R_i R_j)^2 = 1 \text{ } (|i-j| \geq 2) \rangle,$$

whose defining graph is the *orthogonality graph* Γ_n (vertices $0, \dots, n$, edges $|i-j| \geq 2$), and there is a surjection $\rho: W_n \twoheadrightarrow W_T$.

Before computing the envelope we record that the “generic growth” is a well-defined object: it does not depend on which generic shape one picks. Let $F = \ast_{i=0}^n \langle r_i \rangle$ be the free product of $n+1$ copies of $\mathbb{Z}/2$ and let $\rho_\ell: F \rightarrow \text{Isom}(\mathbb{R}^n)$, $r_i \mapsto R_i(\ell)$, be the reflection representation; put $N_{\text{gen}} = \bigcap_{\ell \in U} \ker \rho_\ell$.

Theorem 3 (Generic rigidity). *For every $n \geq 2$ there is a co-null subset $U_{\text{gen}} \subseteq U$ (the complement of a countable union of proper real-algebraic subvarieties) such that $\ker \rho_\ell = N_{\text{gen}}$, hence $W_\ell \cong F/N_{\text{gen}}$, for every $\ell \in U_{\text{gen}}$. Consequently the geodesic growth $s_d(W_\ell)$ is constant on U_{gen} , equal to $s_d(F/N_{\text{gen}})$, and this generic value is the maximum of $s_d(W_\ell)$ over all $\ell \in U$. Moreover U_{gen} contains every ℓ whose coordinates are algebraically independent over $\mathbb{Q}$; in particular it is dense.*

Proof. Each facet normal $m_j(\ell)$ is a fixed polynomial vector in ℓ by Lemma 2, so the reflection $R_j(\ell) = I - 2m_j m_j^\top / (m_j \cdot m_j)$ (with its translation) has entries in $\mathbb{Q}(\ell)$ with denominator the nonvanishing polynomial $m_j \cdot m_j$. Hence every entry of $\rho_\ell(v) = \prod_k R_{i_k}(\ell)$ lies in $\mathbb{Q}(\ell)$, and for each $v \in F$ the locus $V_v = \{\ell \in U : \rho_\ell(v) = I\}$ (the simultaneous vanishing of finitely many rational functions, cleared of their nonvanishing denominators) is Zariski-closed in the irreducible variety U . If $v \notin N_{\text{gen}}$ then $\rho_\ell(v) \neq I$ for some ℓ , so V_v is a *proper* closed subvariety, of measure zero. As F is countable, $B := \bigcup_{v \in F \setminus N_{\text{gen}}} V_v$ is a countable union of proper subvarieties; $U_{\text{gen}} := U \setminus B$ is co-null and dense. For $\ell \in U_{\text{gen}}$ one has $N_{\text{gen}} \subseteq \ker \rho_\ell$ (universal relations hold everywhere) and $\ker \rho_\ell \subseteq N_{\text{gen}}$ (any $v \in \ker \rho_\ell \setminus N_{\text{gen}}$ would place $\ell \in V_v \subseteq B$), so $\ker \rho_\ell = N_{\text{gen}}$ and $W_\ell \cong F/N_{\text{gen}}$. Isomorphic groups with the marked generators have identical geodesic growth, giving the constant value; for any ℓ , $\ker \rho_\ell \supseteq N_{\text{gen}}$ makes W_ℓ a quotient of F/N_{gen} , so $s_d(W_\ell) \leq s_d(F/N_{\text{gen}})$, i.e. the generic value is the maximum. Finally each V_v is defined over $\mathbb{Q}$; if the coordinates of ℓ are algebraically independent over $\mathbb{Q}$ then ℓ lies on no proper $\mathbb{Q}$ -subvariety, so $\ell \notin B$, i.e. $\ell \in U_{\text{gen}}$. Such ℓ exist since $\dim U = n - 1 \geq 1$. $\square$

By Lemma 2 the relations $(R_i R_j)^2 = 1$ for $|i - j| \geq 2$ hold identically in ℓ , so they lie in N_{gen} and F/N_{gen} is a quotient of W_n .

Theorem 4 (The RACG envelope is rational). *Let $c_{\Gamma_n}(x) = \sum_C \text{clique } x^{|C|}$ be the clique polynomial of Γ_n . The spherical growth series of W_n is rational and satisfies*

$$\frac{1}{W_n(t)} = c_{\Gamma_n}\left(\frac{-t}{1+t}\right).$$

Equivalently $W_n(t) = (1+t)^{n+1}/J_{n+1}(t)$, where

$$J_m = (1+t)(J_{m-1} - t J_{m-2}), \quad J_0 = J_1 = 1.$$

One has $J_m = (1+t)^{\lfloor m/2 \rfloor} K_m$ with $K_0 = K_1 = 1$, $K_{2j} = K_{2j-1} - t K_{2j-2}$ and $K_{2j+1} = (1+t)K_{2j} - t K_{2j-1}$, so that

$$W_n(t) = \frac{(1+t)^{\lfloor n/2 \rfloor + 1}}{K_{n+1}(t)}.$$

The reciprocals of the roots of K_{n+1} in $(1/3, 1)$ are the numbers $1 + 2 \cos \frac{2k\pi}{n+3}$, and the growth rate is the largest of them,

$$r_n = 1 + 2 \cos \frac{2\pi}{n+3}.$$

Explicitly

n	$W_n(t)$	r_n
2	$(1+t)^2/(1-t-t^2)$	$\varphi = 1.618\dots$
3	$(1+t)^2/(1-2t)$	2
4	$(1+t)^3/(1-2t-t^2+t^3)$	$1 + 2 \cos \frac{2\pi}{7} = 2.24698\dots$
5	$(1+t)^3/(1-3t+t^2+t^3)$	$1 + \sqrt{2}$
6	$(1+t)^4/(1-3t+3t^3)$	$1 + 2 \cos \frac{2\pi}{9} = 2.53209\dots$
7	$(1+t)^4/(1-4t+3t^2+2t^3-t^4)$	$\varphi^2 = 2.61803\dots$

Proof. W_n is a RACG, so its finite standard parabolics are exactly the cliques of Γ_n , each a group $(\mathbb{Z}/2)^{|C|}$ with growth polynomial $(1+t)^{|C|}$ [3]. Steinberg's formula [4, 1] gives

$$\frac{1}{W_n(t^{-1})} = \sum_{C \text{ clique}} \frac{(-1)^{|C|}}{(1+t)^{|C|}} = c_{\Gamma_n} \left(\frac{-1}{1+t} \right).$$

Substituting $t \mapsto t^{-1}$ and simplifying $-1/(1+t^{-1}) = -t/(1+t)$ yields the displayed identity for $1/W_n(t)$. (The substitution must be carried *inside* the clique polynomial; retaining the argument $-1/(1+t)$ gives a different rational function, already at $n=2$.)

The cliques of Γ_n are the independent sets of its complement, the path P_{n+1} , so $c_{\Gamma_n} = I_{n+1}$, the independence polynomial of the path on $n+1$ vertices, which obeys $I_m = I_{m-1} + xI_{m-2}$, $I_0 = 1$, $I_1 = 1+x$. Put $J_m(t) = (1+t)^m I_m(-t/(1+t))$. Multiplying the recurrence by $(1+t)^m$ gives $J_m = (1+t)J_{m-1} - t(1+t)J_{m-2}$, which is the stated recurrence, with $J_0 = 1$ and $J_1 = (1+t)(1-t/(1+t)) = 1$. By construction $1/W_n(t) = I_{n+1}(-t/(1+t)) = J_{n+1}(t)/(1+t)^{n+1}$. The factorisation $J_m = (1+t)^{\lfloor m/2 \rfloor} K_m$ and the two-step recurrences for K follow by induction on m , splitting on the parity of m ; cancelling $(1+t)^{\lfloor (n+1)/2 \rfloor}$ against $(1+t)^{n+1}$ leaves the exponent $\lceil (n+1)/2 \rceil = \lfloor n/2 \rfloor + 1$.

For the roots, fix $t > 0$ and solve the linear recurrence. Its characteristic equation is $\lambda^2 - (1+t)\lambda + t(1+t) = 0$, of discriminant $(1+t)(1-3t)$.

If $0 < t < 1/3$ both roots are real with $0 < \lambda_- < \lambda_+ < 1$ (their sum is $1+t \leq 4/3$ and their product is $t(1+t) < 4/9$). Writing $J_m = a\lambda_+^m + b\lambda_-^m$, the initial conditions give $a = (1-\lambda_-)/(\lambda_+-\lambda_-) > 0$ and $b = (\lambda_+-1)/(\lambda_+-\lambda_-) < 0$ with $a+b=1$. Then $J_m = \lambda_+^m(a(\lambda_+/\lambda_-)^m + b)$, and the bracket is increasing in m with value 1 at $m=0$, so $J_m > 0$. At $t=1/3$ the root is double and $J_m = (1+m/2)(2/3)^m > 0$. Hence J_m has no root in $(0, 1/3]$.

If $t > 1/3$ the roots are $\lambda_{\pm} = \rho e^{\pm i\theta}$ with $\rho = \sqrt{t(1+t)}$ and $\cos \theta = \sqrt{(1+t)/(4t)}$, so θ ranges over $(0, \pi/3)$ and is strictly increasing in t , because $(1+t)/(4t) = \frac{1}{4} + \frac{1}{4t}$ is strictly decreasing. Then $J_m = \rho^m(\cos m\theta - c \sin m\theta)$, where $J_0 = 1$ is automatic and $J_1 = 1$ forces $c\rho \sin \theta = (t-1)/2$, that is $c = (t-1)/\sqrt{(1+t)(3t-1)}$. Since $\sin^2 \theta = (3t-1)/(4t)$, a direct computation gives $\cot 2\theta = (1-t)/\sqrt{(1+t)(3t-1)} = -c$. Therefore

$$J_m = 0 \iff \cot m\theta = c = -\cot 2\theta \iff \frac{\sin((m+2)\theta)}{\sin m\theta \sin 2\theta} = 0 \iff (m+2)\theta \in \pi\mathbb{Z}.$$

With $m = n+1$ the roots are exactly $\theta = k\pi/(n+3)$, $1 \leq k < (n+3)/3$. Inverting $\cos^2 \theta = (1+t)/(4t)$ gives $t = 1/(4\cos^2 \theta - 1) = 1/(1+2\cos 2\theta)$, so the corresponding t -values are $1/(1+2\cos \frac{2k\pi}{n+3})$. As θ increases with t , the smallest positive root is $k=1$; it exceeds $1/3$ because $1+2\cos \frac{2\pi}{n+3} < 3$, and $1+t > 0$ there, so it is a root of K_{n+1} and not of the stripped factor. Hence $r_n = 1+2\cos \frac{2\pi}{n+3}$. $\square$

Remark 5 (Not a path spectral radius). The number $2\cos\frac{2\pi}{n+3}$ is *not* the spectral radius of the path P_{n+1} , which is $2\cos\frac{\pi}{n+2}$; the two disagree for every $n \geq 2$ (at $n = 2$ they are $0.618\dots$ and $1.414\dots$). The angles produced by the proof are $k\pi/(n+3)$, and the growth rate uses $\cos 2\theta$ at $\theta = \pi/(n+3)$, not $\cos \theta$.

3 The geometric group as a quotient, and what the plane decides

Theorem 1 applied to $\rho: W_n \twoheadrightarrow W_T$ gives an exact alternative, with no genericity or dimension hypothesis.

Proposition 6 (The envelope alternative). *Exactly one of the following holds.*

- (i) ρ is injective. Then $W_T \cong W_n$ and $s_d(W_T) = [t^d]W_n(t)$ for every d ; in particular the geometric growth series is rational with rate r_n .
- (ii) ρ is not injective. Then $n_T := \lceil K(\ker \rho)/2 \rceil < \infty$, $s_d(W_T) = [t^d]W_n(t)$ for $d < n_T$, and $s_{n_T}(W_T) < [t^{n_T}]W_n(t)$.

Proof. If $\ker \rho = \{1\}$ then ρ is an isomorphism of marked groups and (i) is immediate. Otherwise $K(\ker \rho)$ is finite and Theorem 1 with $G = W_n$, $N = \ker \rho$ gives (ii). $\square$

Proposition 7 (The plane). *For $n = 2$ alternative (ii) holds: $\ker \rho \neq \{1\}$.*

Proof. $W_T \leq \text{Isom}(\mathbb{R}^2) = \mathbb{R}^2 \rtimes O(2)$. The group $O(2)$ contains the abelian $SO(2)$ with index 2, so $\text{Isom}(\mathbb{R}^2)$ is virtually solvable *as an abstract group*, hence amenable, and so is every subgroup of it; in particular W_T is amenable. The RACG W_2 is not amenable: its defining graph Γ_2 is not complete multipartite, since its complement P_3 is connected and is not a disjoint union of cliques, and a RACG whose defining graph is not complete multipartite contains a free subgroup of rank 2 [3]. An amenable group is not isomorphic to a non-amenable one, so ρ is not injective. $\square$

Remark 8 (Why this argument stops at $n = 2$). The proof of Proposition 7 does *not* extend to $n \geq 3$, and the natural extension is false. One is tempted to argue that $\text{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$ is amenable because $O(n)$ is compact. Compactness gives amenability of $O(n)$ as a *topological* group, with respect to Haar measure, and that property does not pass to abstract subgroups. As an abstract discrete group $O(n)$ is non-amenable for every $n \geq 3$: already $SO(3)$ contains a free subgroup of rank 2 (Hausdorff; see [5, Thm. 2.1]), which is the standard ingredient of the Banach–Tarski paradox. Hence $\text{Isom}(\mathbb{R}^n)$ is non-amenable as an abstract group for $n \geq 3$ and yields no obstruction at all. At $n = 2$ the argument survives only because $O(2)$ is virtually abelian.

Corollary 9 (The planar horizon). *For $n = 2$ the horizon is $n_T = 10$: $s_d(W_T) = F_{d+3} = [t^d]W_2(t)$ for $1 \leq d \leq 9$ (while $s_0 = 1$ and $F_3 = 2$), and at $d = 10$ the growth deviates, becoming the sequence A396406 with rate $\beta_2 = 1.4916\dots < \varphi = r_2$. That sequence is transcendental over $\mathbb{Q}(x)$ by the companion paper [6]. This is the only dimension in which the deviation has been computed past its onset.*

4 What is proved, and the open problem

Three statements are proved unconditionally and dimension-freely: the universality–deviation dichotomy (Theorem 1); generic rigidity (Theorem 3), which makes “the generic n -orthoscheme growth” a well-defined object $s_d(F/N_{\text{gen}})$ realized by a co-null set of shapes; and the rationality

of the RACG envelope with rate r_n (Theorem 4). To these Proposition 6 adds the exact alternative governing the geometric growth.

Which alternative holds is settled only in the plane, by Proposition 7, where it is (ii) with $n_T = 10$. For $3 \leq n \leq 7$ the geometric growth was computed to depth 13–16 and agrees with the envelope throughout that range. That agreement is consistent with both alternatives: under (i) it would continue forever, and under (ii) it would end at a horizon n_T beyond the computed range. Nothing in the present paper decides between them, and Remark 8 shows that the one soft argument available in the plane is unavailable in higher dimensions.

Question 10 (The alternative in higher dimensions). For $n \geq 3$, is $\rho: W_n \rightarrow W_T$ injective for the generic n -orthoscheme? Equivalently, by Proposition 6, is the geometric growth series equal to the rational envelope $W_n(t)$ in every degree? If ρ is injective for all $n \geq 3$ then the plane is the unique dimension with non-rational orbit growth. If instead ρ has nontrivial kernel in some dimension $n \geq 3$, then the growth deviates below the envelope at $n_T = \lceil K(\ker \rho)/2 \rceil$, and the arithmetic of the deviated series past that horizon is open even in the cases where the horizon can be located.

The companion papers are recast accordingly: their object, the planar right-triangle group, is the one dimension in which the quotient of the Coxeter envelope has been shown proper and then carried past its deviation [6, 7]. Whether that behaviour is two-dimensional or the first instance of a general phenomenon is Question 10.

Certification

Theorem 4 is certified by `code/zeta_probe/tools/racg_envelope` (Rust, exact integer polynomial arithmetic), which for $2 \leq n \leq 30$ checks that the recurrence for J_m and the Steinberg expansion agree coefficient by coefficient, performs the division by $(1+t)^{\lfloor (n+1)/2 \rfloor}$ exactly, and matches the smallest positive root of K_{n+1} against $1/(1+2\cos \frac{2\pi}{n+3})$. Independently, for $2 \leq n \leq 6$ it enumerates the group by breadth-first search in the canonical geometric representation, whose matrices are integral here because the bilinear form takes only the values 1, 0, -1 , and which is faithful by Tits’ theorem [1]; the sphere sizes agree with the series coefficients to depths 9 to 14. The same program exhibits the two corrections recorded above: the substitution $-1/(1+t)$ in place of $-t/(1+t)$ changes the rational function already at $n = 2$, and the tabulated $n = 7$ denominator of an earlier version, $1 - 2t - 3t^2 + 4t^3 - t^4$, is the negated reciprocal of the correct $1 - 4t + 3t^2 + 2t^3 - t^4$ and predicts 6 elements at depth 1 in a group with 8 generators.

The arithmetic of Theorem 1 and the root identity of Theorem 4 are machine-checked in Lean 4 in `lean/with_mathlib/EnvelopeDichotomy.lean` (twelve theorems, no `sorry`), and Lemma 2 in `lean/with_mathlib/OrthoschemeNormals.lean` (seven theorems, no `sorry`): the support pattern `supp_disjoint`, the vanishing of the inner product on disjoint supports `sum_eq_zero_of_disjoint`, and the three adjacent inner products, which are nonzero for every positive leg tuple. That file is what replaces the earlier “verified symbolically for $n \leq 5$ ”.

References

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